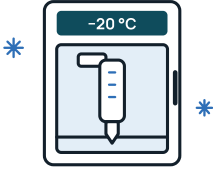


# Exhaled Breath Condensate (EBC) Collection

Five steps from the pre-cooled condenser to the analysis laboratory, for nurses and physicians collecting samples in the breath phenotyping study.

**Cold chain -20 °C**  
unbroken from collection to lab

1



### Retrieve the pre-cooled condenser

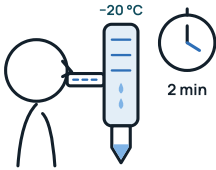
Take the -20 °C breath condenser from the freezer in the sample collection room. Check that the Eppendorf tube is seated at the bottom of the condensation tube, fit a clean mouthpiece and start at once while the tube is cold.

✓ Gloves on. Label the tube: participant ID · date · time.

-20 °C

PRE-COOLED CONDENSER

2



### Exhale strongly for 2 minutes

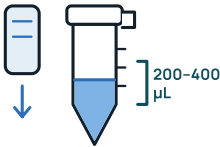
The participant breathes in through the nose and exhales strongly and steadily through the mouthpiece into the -20 °C tube for 2 timed minutes. Keep the device upright so the condensate runs down into the Eppendorf tube.

✓ No coughing, talking or saliva into the mouthpiece.

2 min

STRONG, STEADY EXHALATION

3



### Recover the condensed sample

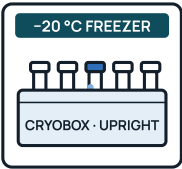
Detach the Eppendorf tube from the bottom of the condenser. Expected yield is about 200–400 µL of condensate, which may be partly frozen. Cap the tube immediately and record the approximate volume on the sample form.

✓ Low volume? Record it. Never top up or pool samples.

200–400 µL

EXPECTED YIELD

4



### Store the Eppendorf tube at -20 °C

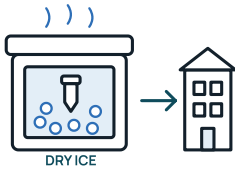
Place the capped, labelled tube upright in the cryobox in the -20 °C freezer immediately after collection and log its position and time. Return the condenser to the freezer to re-cool for the next participant.

✓ Once frozen, it stays frozen: no thaw and refreeze.

-20 °C

STORE AT ONCE, UPRIGHT

5



### Ship to Gipfel Life Sciences on dry ice

Send frozen samples to the Gipfel Life Sciences laboratory in an insulated box packed with dry ice so they stay at -20 °C or colder throughout. Enclose the sample manifest (ID, date, volume) and notify the lab before dispatch.

✓ Samples must never thaw in transit.

Dry ice

-20 °C OR COLDER IN TRANSIT

### CRITICAL POINTS

- ✱ **Cold chain.** Condenser pre-cooled, tube frozen at once, dry ice in transit.
- 🧼 **Clean sample.** New mouthpiece each time; no saliva, no coughing.
- 📄 **Traceability.** Label and log every tube: ID, date, time, volume.

### RECORD ON THE SAMPLE FORM

|                      |                      |
|----------------------|----------------------|
| PARTICIPANT ID       | DATE · START TIME    |
| <input type="text"/> | <input type="text"/> |
| VOLUME, µL           | POSITION · INITIALS  |
| <input type="text"/> | <input type="text"/> |